

Battery Chemistry Research and Comparison

Types

Technology	Major producers (2023)	Target application	Advantages
Lithium nickel manganese cobalt oxide NMC , $\text{LiNi}_x\text{Mn}_y\text{Co}_z\text{O}_2$	Ronbay Technology, Easpring, Ecopro, Umicore, L&F, Posco ^[96]	Electric vehicles, power tools, grid energy storage	Good specific energy and specific power density
Lithium nickel cobalt aluminium oxide NCA , LiNiCoAlO_2	Ronbay Technology, Ecopro ^[96]	Electric vehicles, power tools, grid energy storage	High energy density, good life span
Lithium nickel cobalt manganese aluminium oxide NCMA , $\text{LiNi}_{0.89}\text{Co}_{0.05}\text{Mn}_{0.05}\text{Al}_{0.01}\text{O}_2$	LG Chem, ^[97] Hanyang University ^[98]	Electric vehicles, grid energy storage	Good specific energy, improved long-term cycling stability, faster charging
Lithium manganese oxide LMO , LiMn_2O_4	Posco, L&F ^[96]	Power tools, electric vehicles ^[99]	Fast charging speed, cheap
Lithium iron phosphate LFP , LiFePO_4	Shenzhen Dynanonic, Hunan Yuneng, LOPAL, Ronbay Technology ^[96]	Electric vehicles, ^[95] grid energy storage ^[94]	Higher safety compared to layered oxides. Very long cycle life. Thermal stability >60 °C (140 °F)
Lithium cobalt oxide LCO , LiCoO_2	Easpring, Umicore ^[96]	Handheld electronics ^[96]	High energy density

Figure 1: Table of Positive Electrode Types

Following our Facility Advisor, We'll be only conducting further research into [Lithium Nickel Manganese Cobalt Oxide \(NMC\)](#) and [Lithium Iron Phosphate \(LFP\)](#).

Properties

NMC

- Batteries with NMC Cathode are 3.6-3.7 V
- High Energy storage
- Moderate safety
- Operates under extreme temperature conditions
- Batteries with extra nickel are good for EVs

LFP Deep Dive

Lithium Iron Phosphate (LFP) is a lithium-ion battery chemistry that uses an iron-phosphate cathode, notable for its strong thermal stability and long service life. LFP cells typically operate at a lower nominal voltage (~3.2 V per cell) and exhibit a very flat discharge voltage profile, meaning cell voltage changes little across much of the usable state-of-charge range. This chemistry is widely used in stationary energy storage, buses, and safety-critical EV platforms where durability and abuse tolerance are prioritized.

For our Formula SAE Electric application, LFP's primary strengths are its reduced thermal runaway risk and forgiving thermal behavior, which can lower safety risk during early development and testing. However, its lower energy density and lower cell voltage result in heavier battery packs and higher series cell counts for a given pack voltage, negatively impacting vehicle mass, packaging, and performance. Additionally, the flat voltage profile complicates state-of-charge estimation, increasing reliance on accurate current sensing and BMS algorithms.

Technical Advantages

- **Intrinsic Safety and Thermal Stability:** LFP cells are widely regarded as one of the safest lithium-ion chemistries due to their strong resistance to thermal runaway. In practical terms, LFP cells can tolerate higher temperatures and certain abuse conditions with a lower likelihood of catastrophic failure compared to nickel-based chemistries. Thermal runaway onset temperatures for LFP cells are typically above 250-270 degrees Celsius, which provides an increased safety margin during testing, integration, and fault scenarios. For our Formula SAE Electric application, this improves fault tolerance during early development and reduces the severity of potential thermal incidents.
- **Long Life Cycle and Durability:** LFP chemistry is known for its long cycle life, with commercial cells commonly rated for thousands of full charge-discharge cycles before significant capacity degradation occurs. This durability allows repeated testing, validation, and tuning without rapid battery degradation, making LFP attractive for development-heavy projects or teams expecting frequent charge cycles throughout a season.
- **Cost and Material Availability:** LFP cells rely on iron and phosphate rather than cobalt or high-nickel content materials. As a result, LFP generally offers a lower cost per kilowatt-hour and reduced exposure to supply chain volatility. This can be beneficial for student teams operating under budget constraints or seeking easily replaceable cells for long-term development.
- **High Current Capability:** Modern LFP cells are capable of supporting relatively high continuous and peak discharge currents, making them suitable for electric propulsion applications. Many commercially available LFP cells can meet the power demands of a Formula SAE drivetrain, particularly in short acceleration events, assuming that the pack is sized appropriately.

Technical Challenges

- **Lower Energy Density and Increased Pack Mass:** Compared to NMC, LFP cells have significantly lower gravimetric and volumetric energy density. This means that an LFP battery pack designed to meet the same energy requirement will generally be heavier and larger. In a

Formula SAE Electric racecar, increased accumulator mass negatively impacts acceleration, handling, and overall competitiveness, making energy density a critical drawback of LFP.

- **Flat Voltage Profile and SOC Estimation Challenges:** LFP cells exhibit a very flat voltage curve across most of their usable state-of-charge range. While this behavior is fine from a power delivery standpoint, it makes voltage-based SOC estimation unreliable. As a result, a BMS must rely more heavily on accurate current measurement and coulomb counting, increasing software complexity and sensitivity to sensor drift.
- **Higher Series Cell Count for High-Voltage Packs:** Due to LFP’s lower nominal voltage, achieving a typical Formula SAE pack voltage in the 300-400 V range requires a larger number of cells connected in series compared to NMC. This increases pack complexity, sensing channel count, and BMS balancing requirements, and can complicate mechanical packaging and segmentation strategies.

Metric	LFP Typical Value	Relevance to Application
Gravimetric Energy Density	120-160 Wh/kg	Determines the minimum pack mass for a target total energy, directly influencing vehicle weight distribution.
Volumetric Energy Density	300-350 Wh/L	Dictates the physical volume of the pack, impacting chassis integration, center of gravity, and bodywork.
Nominal Cell Voltage	3.2-3.3 V	Defines the series cell count for a target pack voltage. A higher count increases BMS channel requirements.
Operational Voltage Window	2.5-3.6 V	Sets the upper and lower limits for BMS voltage protection. The relatively low upper

		limit reduces stress on cell-level components.
Cycle Life (to 80% capacity)	2,000-5,000 cycles	Informs the pack's usable lifespan and long-term value, supporting reuse across multiple development and competition seasons.
Max Continuous Discharge Rate	3-5C	Must be validated against the motor/inverter's peak current demand to ensure the pack can deliver required power without excessive voltage sag.
Thermal Runaway Onset Temperature	Greater than 250-270 degrees Celsius	Influences the design requirements for the thermal management system and containment strategy, potentially allowing for simpler solutions.
Approximate Cell Cost	\$80-\$120 / kWh	A primary factor in overall accumulator budget, affecting the feasible pack size or allocation of funds to other vehicle subsystems.

NMC Deep Dive

Nickel Manganese Cobalt Oxide (NMC) is a family of lithium-ion chemistries that combine nickel, manganese, and cobalt in the cathode to optimize energy density, power capability, and cost. NMC cells typically have a higher nominal voltage (~3.6-3.7 V per cell) and a more sloped discharge voltage curve, which provides more usable voltage information for monitoring and control. This chemistry is dominant in modern electric vehicles and high-performance battery systems.

For our Formula SAE Electric application, NMC's key advantage is its high specific energy, enabling lighter and more compact battery packs that improve acceleration, handling, and overall vehicle competitiveness. The higher operating voltage reduces current for a given power level, simplifying conductor sizing and lowering resistive losses. The primary tradeoff is reduced inherent thermal stability compared to LFP, which demands stricter thermal monitoring, conservative protection thresholds, and a robust BMS to manage safety under high-power racing conditions.

Technical Advantages

- **High Energy Density Enabling Lightweight Packs:** NMC chemistry offers some of the highest gravimetric and volumetric energy density among commercially available lithium-ion batteries. This allows a battery pack to store the required energy while minimizing total mass and volume. In a Formula SAE Electric racecar, this directly improves acceleration, handling, and overall competitiveness by reducing accumulator weight and easing packaging constraints within the chassis.
- **Favorable Voltage Profile for Monitoring and Control:** NMC cells exhibit a relatively sloped voltage curve across much of their usable state-of-charge range. This makes cell voltage a useful indicator of SOC when combined with current-based estimation methods. For BMS design, this simplifies initial SOC estimation, diagnostics, and fault detection compared to flatter-voltage chemistries.
- **High Power Capability:** NMC cells are capable of delivering very high continuous and peak discharge currents, making them well suited for the aggressive acceleration and regenerative braking demands of a Formula SAE drivetrain. Power-optimized NMC cells can reliably support high transient loads while maintaining stable voltage behavior.

Technical Challenges

- **Reduced Thermal Stability:** NMC chemistry is inherently less thermally stable than LFP and requires more careful thermal and safety management. Under abusive conditions, NMC cells can enter thermal runaway at lower temperatures, making robust thermal monitoring,

conservative operating limits, and reliable shutdown mechanisms essential. For a Formula SAE application, this places greater importance on BMS reliability, sensor coverage, and fault response logic.

- **Shorter Cycle Life:** NMC cells typically experience faster capacity degradation than LFP, particularly when operated at high voltages, high temperatures, or high C-rates. While this is generally acceptable for a race-focused application with a limited service life, it reinforces the need for conservative voltage limits and careful charge control within the BMS to preserve pack health over a competition season.
- **Higher Cost and Material Sensitivity:** NMC cathodes contain cobalt and nickel, which contribute to higher material cost and supply chain sensitivity compared to iron-based chemistries. For student teams, this can increase upfront costs and make cell replacement more expensive. However, this tradeoff is often accepted in exchange for improved performance and reduced pack mass.

Metric	NMC Typical Value	Relevance to Application
Gravimetric Energy Density	180-250+ Wh/kg	Enables a lighter pack for a target energy, a critical advantage for vehicle dynamics.
Volumetric Energy Density	600-750 Wh/L	Allows for a more compact pack, easing integration into the vehicle's frame and improving design flexibility for aerodynamics and weight distribution.
Nominal Cell Voltage	3.6-3.7 V	Reduces the series cell count for a target pack voltage. A lower count simplifies BMS channel requirements and wiring.

Operational Voltage Window	3.0-4.2 V	Defines a wider absolute range but requires precise control at the top end. The BMS must strictly enforce the upper limit to preserve cycle life.
Cycle Life (to 80% capacity)	1,000-2,000 cycles	Defines the pack's service life. May necessitate more frequent pack refurbishment or replacement over the lifetime of the vehicle program.
Max Continuous Discharge Rate	3-5C+ (cell dependent)	Must be verified against peak motor current. High-power NMC variants readily meet racing demands.
Thermal Runaway Onset Temp.	150-200 degrees Celsius	Dictates a non-negotiable requirement for an effective thermal management system such as liquid cooling and robust fault detection.
Approximate Cell Cost	\$100-\$150+ / kWh	Impacts the total accumulator budget, potentially limiting pack size or requiring greater funding allocation.